

smalltouch

smalltouch develops and produces a number of components within the area of GNSS. The American NAVSTAR GPS is such a system. The components are among other things used for measuring of road systems, generating data for maps (GIS) or monitoring of larger building constructions.

The components often represent a centre in larger systems where they handle acquisition and storing of measured data for further processing. Furthermore smalltouch's touch screen makes up the interface to the user.

On enquiry smalltouch develops test tools i.e. for accelerations and stress test of equipment that are part of land mobile solutions.

smallPOS



In combination with a GPS receiver the smallPOS (small Orientation & Positioning System) will deliver continuous navigation information even in areas with poor or non GPS satellite view. The smallPOS' output is standard NMEA position-strings and raw real-time heading pitch and roll information.

The output is generated on the basis of data from sensors such as gyro, inclinometers of high quality (if used in a car an odometer is connected) and the GPS receiver.

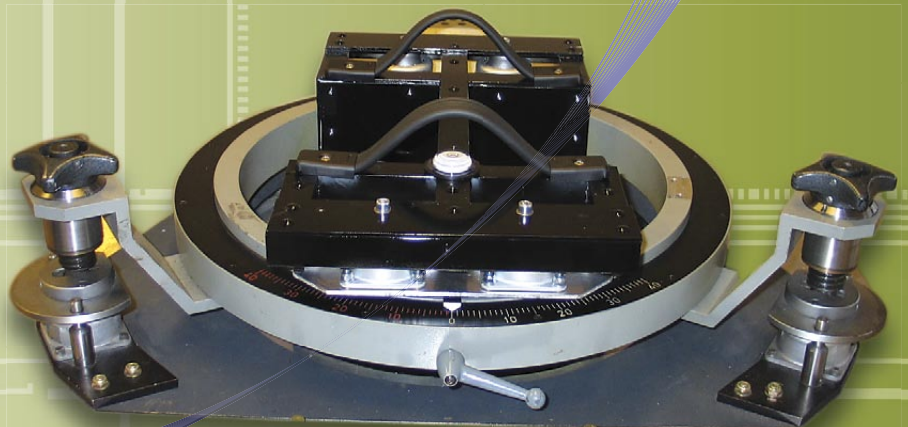
Due to the inclinometer (pitch information) the smallPOS unit produces three-dimensional map information.

A wide range of GPS receivers (with or without DGPS/RTK) from producers such as NovAtel, Trimble Ashtec etc. can be used and controlled by the smallPOS. The smallPOS is also produced in a version with internal GPS receiver.

Applications:

smallPOS provides orientation information for applications beyond those served by GPS alone. Application developers use aided inertial position and orientation data for an ever-growing list, including:

- Rapid data collection for direct entry to GIS databases
- Road track geometry horizontal and vertical
- Intelligent vehicle highway system applications
- Pavement management data collection
- Corridor surveys (video logging)
- Risk Management
- Civil engineering
- Motion compensation for vehicle-mounted sensors (e.g. antenna, video or still camera).



This example shows smallPOS units implemented in a cameramount for aerial survey.

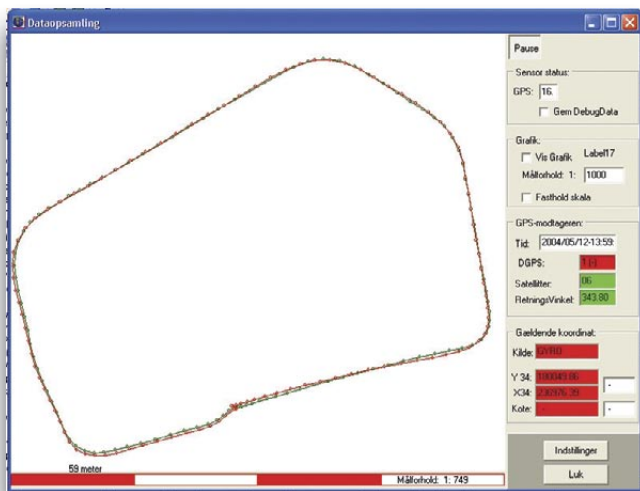


Product case:

Torben Weinkouff Rasmussen from The National Road Directorate in Denmark explains:

"We measured 135 km of busy highway near Copenhagen on three evenings/nights. And because the speed when measuring was 20-30 km/h, we were able to avoid large road closings.

This was possible only due to the survey vehicle developed in co-operation with smalltouch. The smallIPOS is part of this survey system. A PC gathers information from GPS odometer, gyro, tilt and compass sensors, which is vital information for us."



This is a test example from The National Road Directorate in Denmark. On a road section of approximately 700 metres a clockwise lap is made (duration approximately 110 seconds). On the entire lap the GPS conditions are generally good. On the first lap (illustrated by the green line) a Trimble 4700 GPS-receiver using RTK is used for measuring. On the second lap the GPS-receiver is disconnected, and the smallIPOS generates position information (illustrated by the red line). The maximum deviation is four metres.

Henrik Vad Jensen from National Road Directorate in Denmark says:

"Compared to the price smallIPOS delivers accuracy. For quite some time now The National Road Directorate in Denmark has inquired for a product for our inspection vehicles, which makes positioning possible in areas where GPS positioning is not possible. The sensors of the smallIPOS unit make it possible to make this positioning - with reasonable accuracy. Furthermore the fact that it is possible to extract raw sensor data from the smallIPOS unit creates a unique chance for optimizing the accuracy as to the dynamic area, in which the unit is used, with your own algorithm. The close link between the GPS receiver and the smallIPOS unit ensures a precise timing between all sensors contributing to the positioning calculation. The smallIPOS unit is a nice flexible positioning system, for which we have high expectations."



With a physical measurement of 85x100x40 mm and a power consumption of less than 2W (12VDC) the smallIPOS unit has an output of up to 40 raw heading pitch and roll data. The heading tolerance is less than 0.5% of FS and the pitch roll information better than 0.1 angular degree. This applies within temperatures of -20°C to +70°C.

For further information please refer to www.smalltouch.dk:

- smallIPOS LM technical specifications.pdf
- smallIPOS LM connector and configuration.pdf
- smallIPOS LM with Ashtech DG16.pdf
- smallIPOS LM with Trimble BD950.pdf
- smallIPOS port extension cable.pdf
- smallIPOS Enclosure.pdf